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MAJOR PROJECT PHASE-I (BAI685) REPORT

on

“Explainable AI-Driven Crop Advisory Framework for Precision Agriculture”

Submitted in the partial fulfilment of requirements for the award of Degree

B.E. in Artificial Intelligence & Machine Learning

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The work presented in this synopsis is original and has not been submitted elsewhere for the award of any degree.

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The work is being carried out under my supervision and guidance. The contents of this synopsis have not been submitted to any other institution or university for the award of any degree.

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DECLARATION

We hereby declare that the synopsis entitled “**Explainable AI-Driven Crop Advisory Framework for Precision Agriculture**” submitted in partial fulfilment of the requirements for the award of the degree of **Bachelor of Engineering in Artificial Intelligence and Machine Learning**, is a record of the work being carried out by us under the supervision of **Prof. Kotramma T S** .

The work presented in this synopsis is original and has not been submitted, either in full or in part, for the award of any other degree in this or any other Institution or University.

We further declare that, in accordance with ethical standards in academic reporting, proper acknowledgements have been made wherever the contributions of others have been referenced.

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ABSTRACT

The **Explainable AI-Driven Crop Advisory Framework for Precision Agriculture** is an advanced, data-driven solution aimed at enhancing agricultural productivity by assisting farmers in making informed decisions. This system leverages Machine Learning algorithms to analyze multiple factors such as soil composition, weather conditions, rainfall patterns, temperature variations, and historical crop yield data to recommend the most suitable crops for cultivation. A key feature of this system is the integration of Explainable AI (XAI), which ensures that the predictions made by the model are transparent and interpretable, allowing farmers to understand the reasoning behind each recommendation rather than treating the system as a black box. The system can incorporate real-time data through IoT sensors to monitor soil moisture, nutrient levels, and environmental conditions, thereby improving prediction accuracy and adaptability. Additionally, AI agents can be integrated to provide personalized recommendations, timely alerts, and dynamic decision support based on changing agricultural conditions. The platform may also offer guidance on fertilizer usage, irrigation scheduling, and early detection of pests or diseases, contributing to efficient resource management and risk reduction. Designed with a user-friendly interface accessible via mobile or web applications, the system ensures ease of use even for farmers with limited technical expertise. This project is highly beneficial for AIML students as it combines key concepts such as data preprocessing, supervised learning, model evaluation, and interpretability while addressing real-world challenges in agriculture, ultimately promoting sustainable farming practices and improved crop yield.

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Chapter 1

INTRODUCTION

The rapid advancement of Artificial Intelligence (AI) and data-driven technologies has significantly transformed the agricultural sector, enabling farmers to make informed decisions regarding crop selection, irrigation, and disease management. However, despite these advancements, farmers still face challenges due to unpredictable weather conditions, lack of expert guidance, and limited access to real-time agricultural insights. Most existing systems provide recommendations without proper explanations, making it difficult for farmers to trust and adopt these technologies.

To address these challenges, the proposed project, “Smart Crop Advisory System with Explainable AI for Farmer Decision Support,” introduces an AI-driven solution that integrates Machine Learning (ML) and Explainable AI (XAI) techniques. The system analyzes agricultural data such as soil properties, weather conditions, and crop images to generate accurate recommendations. Unlike traditional systems, it also provides clear explanations for its predictions, improving transparency and usability.

The system consists of multiple intelligent modules, including crop recommendation, disease detection, irrigation prediction, and an explanation module. These components work together to provide personalized, accurate, and interpretable advisory services. By combining predictive analytics with explainability, the system enhances decision-making and promotes trust among farmers.

The Smart Crop Advisory System with Explainable AI for Farmer Decision Support is an innovative application of Artificial Intelligence and Machine Learning aimed at transforming traditional farming practices into data-driven decision-making processes. Agriculture plays a vital role in economies like India, where farmers often rely on experience and uncertain environmental conditions to choose crops, leading to inconsistent yields and financial risks. This project introduces a smart system that utilizes Machine Learning models to analyze various

parameters such as soil nutrients, temperature, rainfall, humidity, and historical agricultural data to recommend the most suitable crops for cultivation.

1.1 Background Of The Problem

In recent years, Artificial Intelligence (AI) and Machine Learning (ML) have emerged as powerful tools for transforming agriculture into a data-driven industry. AI-based systems can analyze large volumes of agricultural data, including soil properties, weather conditions, and historical crop patterns, to provide accurate recommendations. Studies have shown that machine learning algorithms such as Random Forest and Decision Trees can significantly improve crop prediction accuracy and resource optimization [1], [3].

However, one of the major limitations of existing AI-based agricultural systems is the lack of transparency in decision-making. Most models act as “black boxes,” providing predictions without explaining the reasoning behind them. This lack of interpretability reduces trust among farmers, who may be hesitant to rely on automated recommendations [2].

1.2 Overview of Smart Crop Advisory System

A Smart Crop Advisory System is an intelligent decision-support system designed to assist farmers in making informed agricultural decisions. It integrates multiple data sources such as soil nutrient levels (Nitrogen, Phosphorus, Potassium), pH value, temperature, humidity, and rainfall to recommend the most suitable crops and farming practices.

The proposed system utilizes machine learning algorithms to analyze input data and generate recommendations. Additionally, Explainable AI techniques such as SHAP (Shapley Additive Explanations) and LIME (Local Interpretable Model-Agnostic Explanations) are incorporated to provide insights into how each input feature contributes to the final prediction [2], [13]. This transparency helps farmers understand the reasoning behind recommendations, thereby increasing trust and usability.

1.3 Motivation And Gap Identified

Despite significant advancements in agricultural technology, most existing crop advisory systems continue to provide static and generalized recommendations that do not adapt to dynamic environmental conditions. These systems often rely on predefined datasets and fail to incorporate real-time factors such as changing weather patterns, soil variability, and seasonal fluctuations, which are critical for accurate agricultural decision-making. As a result, the recommendations generated may not be suitable for current field conditions, leading to reduced crop productivity and inefficient resource utilization. A Smart Crop Advisory System integrated with Explainable AI can address these issues by providing accurate recommendations along with clear explanations, thereby enhancing farmer trust, improving decision-making, and promoting sustainable agricultural practices.

1.4 Scope of the Project

This project is to develop a Smart Crop Advisory System using machine learning and Explainable AI to assist farmers in making better agricultural decisions. The system analyzes soil parameters such as nitrogen, phosphorus, potassium, and pH along with environmental factors like temperature, humidity, and rainfall. It provides suitable crop recommendations based on these inputs. Additionally, the system suggests proper fertilizer usage and irrigation planning to improve productivity. A key feature of the project is the use of Explainable AI, which helps in providing clear reasons for the recommendations. This enhances transparency and builds trust among farmers. The system is designed with a user-friendly interface for easy accessibility. It can be further extended by integrating real-time weather data and IoT sensors. Future enhancements may include pest detection and multilingual support. Overall, the project aims to create a scalable and intelligent agricultural decision-support system.

1.5 Importance of The Project

This project plays a significant role in improving agricultural productivity by providing farmers with accurate, data-driven recommendations based on soil and environmental conditions. It helps in reducing risks associated with improper crop selection, inefficient irrigation, and excessive fertilizer usage. By leveraging machine learning techniques, the system enables better planning and decision-making, leading to higher crop yield and optimized resource utilization. A key advantage of the project is the integration of Explainable AI, which provides clear and understandable reasons for each recommendation. This transparency increases farmer confidence and encourages the adoption of AI-based solutions in agriculture.

Chapter 2

LITERATURE SURVEY

In recent years, the application of Artificial Intelligence (AI) and Machine Learning (ML) in agriculture has gained significant attention due to its potential to improve productivity and decision-making. AI-based crop recommendation systems utilize historical data, soil parameters, and climatic conditions to suggest suitable crops. Studies have shown that machine learning algorithms such as Decision Trees, Random Forest, and Support Vector Machines provide high accuracy in crop prediction tasks [1], [2].

Deep learning techniques have also been widely used in precision agriculture for yield prediction and disease detection. According to Kamilaris and Prenafeta-Boldú [3], deep learning models can analyze large-scale agricultural datasets and extract meaningful patterns to support decision-making. Similarly, Khaki and Wang [4] demonstrated that deep neural networks improve crop yield prediction by incorporating environmental and temporal data.

However, despite these advancements, most existing systems lack transparency and interpretability. Machine learning models often function as black boxes, making it difficult for users to understand how predictions are generated. Ribeiro et al. [6] introduced LIME, a technique that explains model predictions locally, while Lundberg and Lee [7] proposed SHAP, which provides a unified approach to interpreting model outputs. These Explainable AI (XAI) techniques have been widely adopted to improve trust and usability of AI systems.

In the agricultural domain, the use of Explainable AI is still emerging. Recent studies indicate that integrating XAI with crop recommendation systems can significantly enhance farmer trust and adoption [8]. Additionally, research by FAO highlights the importance of combining AI with sustainable farming practices to ensure long-term agricultural development [9].

Sl. No	Title	Year	Methods Used	Accuracy / Outcome	Gap Found
1.	LSTM-based Irrigation Decision System	2022	LSTM, Predictive Control	Efficient water usage ($R^2 \approx 0.82$)	Black-box model; lacks transparency
2.	AI-based Crop Advisory with Ensemble Learning	2023	Ensemble ML, IoT + Static Data	Accurate crop prediction	Limited user-friendly explanation support
3.	AgroXAI: Explainable Crop Recommendation System	2024	ML + SHAP, LIME, ELI5	Provides local & global explanations	Requires large datasets and infrastructure
4.	Explainable AI for Precision Agriculture Decision-Making	2025	CNN + Grad-CAM, LIME	High interpretability with accurate predictions	Mainly focused on disease detection, not full advisory system
5.	Smart Crop Advisory Systems using AI & ML	2025	Machine Learning, Data Analytics	Improved productivity & decision-making	Lack of personalized and transparent recommendations

2.2 Summary of The Literature Review

The literature survey indicates that AI and ML techniques have significantly improved agricultural decision-making by providing accurate crop recommendations and yield predictions. IoT-based systems enable real-time monitoring of environmental conditions, enhancing precision farming practices. Explainable AI techniques such as LIME and SHAP have addressed the issue of transparency in machine learning models by providing interpretable insights.

However, most existing systems suffer from several limitations. They lack integration of multiple data sources such as soil, weather, and historical data. Many systems provide static

recommendations without adapting to real-time conditions. Additionally, the use of Explainable AI in agriculture is still limited, and user-friendly interfaces for farmers are often lacking. Therefore, there is a need for a comprehensive system that integrates machine learning, real-time data processing, and Explainable AI to provide accurate, transparent, and adaptive agricultural recommendations.

2.3 Problem Statement

Farmers face multiple challenges such as unpredictable weather, varying soil conditions, pest attacks, and lack of timely expert guidance, which often lead to poor decision-making and reduced crop productivity. Many farmers still rely on traditional knowledge or limited information that may not be suitable under changing environmental conditions. Although modern technologies like Artificial Intelligence (AI) and Machine Learning (ML) can provide accurate crop recommendations and predictions, most of these systems function as “black boxes,” making it difficult for farmers to understand how decisions are made and reducing their trust in such solutions. Therefore, there is a need for a Smart Crop Advisory System that delivers accurate, data-driven recommendations for crop selection, irrigation, fertilization, and pest control. By integrating soil data, weather information, and historical crop patterns, and incorporating Explainable AI (XAI) techniques to present clear and simple explanations, the system can improve decision-making, increase productivity, reduce risks, and encourage farmers to adopt AI-based technologies.

2.4 Objectives

- To collect and preprocess soil, weather, and agricultural data for crop analysis.
- To develop a machine learning-based crop advisory system integrated with Explainable AI for transparent recommendations.
- To develop the system and provide a user-friendly decision support tool to assist farmers in selecting suitable crops.
- To proposed models are analyzed using performance metrics such as Accuracy, Precision, Recall, F1-Score, and Confusion Matrix.

Chapter 3

METHODOLOGY

3.1 Description Of Methodology Components

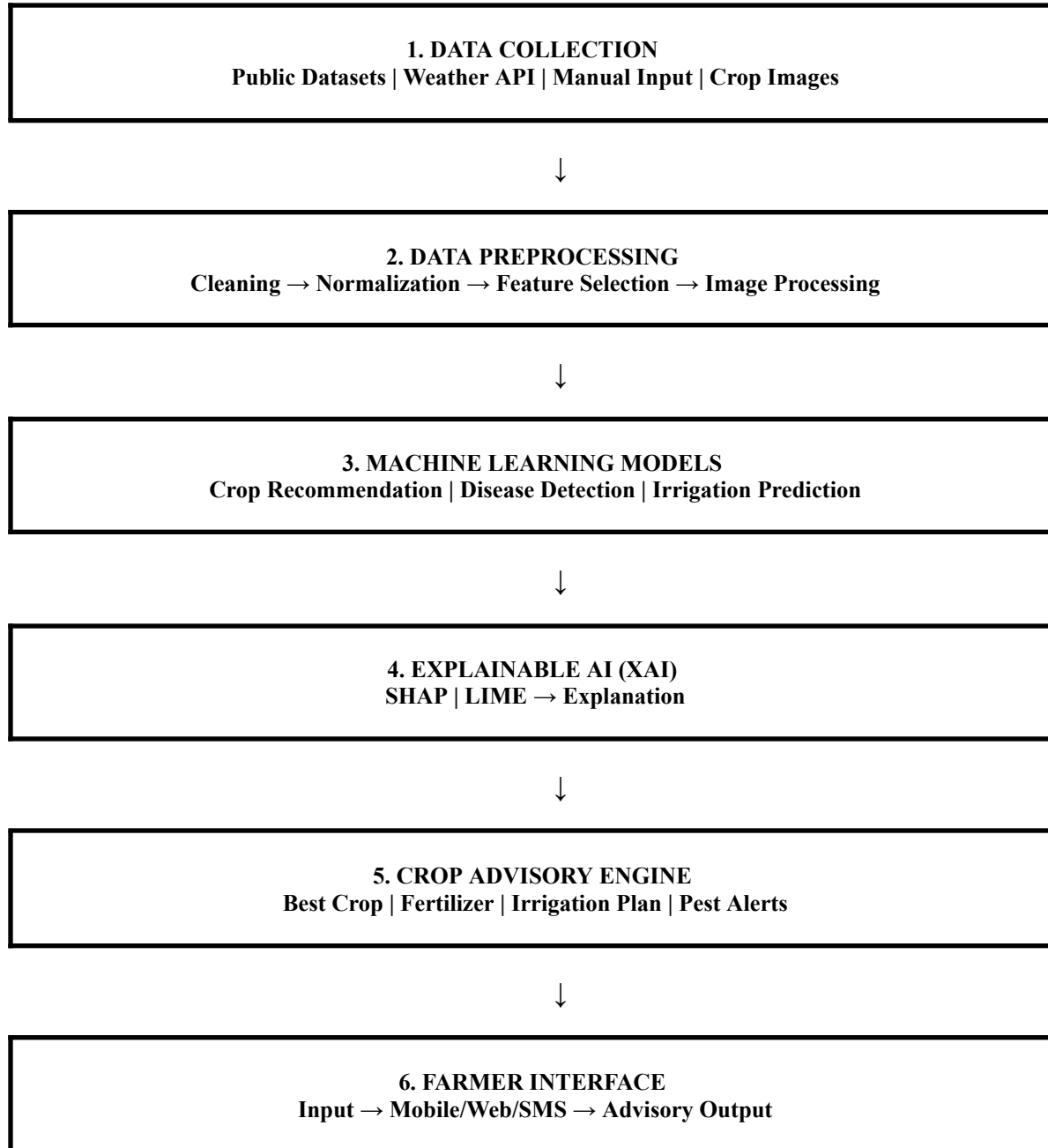


Fig 3.1: Methodology workflow for Explainable AI-Driven Crop Advisory Framework for Precision Agriculture

- **Data Collection**

The first step involves gathering data from multiple sources. This includes public datasets containing information about crops, soil properties, and yield patterns. Real-time environmental data such as temperature, humidity, and rainfall is obtained using weather APIs. The system may also accept crop images uploaded by users for disease detection. This multi-source data collection ensures comprehensive input for better decision-making.

- **Data Preprocessing**

The collected data is cleaned and prepared for analysis. Missing values are handled, and irrelevant or inconsistent data is removed. The data is then normalized and encoded to ensure uniformity across different features. Feature selection techniques are applied to identify the most important parameters influencing crop prediction. For image data, resizing and filtering techniques are used to enhance quality before feeding it into machine learning models.

- **Machine Learning Models**

In this stage, different machine learning models are used for various agricultural predictions:

- **Crop Recommendation:** Algorithms like Random Forest and XGBoost are used to suggest the best crop based on soil and weather conditions.
- **Disease Detection:** Convolutional Neural Networks (CNN) analyze crop images to detect diseases.
- **Irrigation Prediction:** Regression models estimate water requirements based on environmental factors.

These models are trained on historical datasets to improve accuracy and reliability.

- **Explainable AI (XAI)**

To overcome the black-box nature of machine learning models, Explainable AI techniques such as SHAP and LIME are used. These methods provide clear explanations for predictions by showing how each input feature contributes to the final result. For example, the system may explain that a particular crop is recommended due to high

nitrogen levels and suitable rainfall. This transparency increases user trust and understanding.

- **Crop Advisory Engine**

The advisory engine integrates outputs from all models to generate actionable recommendations. It provides:

- Best crop selection.
- Fertilizer dosage suggestions.
- Irrigation planning.
- Pest and disease alerts along with remedies.

This module acts as the core decision-making unit of the system.

- **Farmer Interface**

The final results are presented to farmers through a user-friendly interface such as a mobile application, web portal. Farmers can input their data and receive clear, easy-to-understand recommendations along with explanations. This ensures accessibility even for users with limited technical knowledge.

3.2 Working Flow Of The System

The system begins by collecting input data from users and external sources such as datasets and weather APIs. The collected data is preprocessed to remove inconsistencies and prepare it for analysis. Machine learning models then process the data to generate predictions related to crop selection, disease detection, and irrigation requirements. Explainable AI techniques are applied to interpret the results and provide meaningful explanations. The advisory engine combines all outputs and generates final recommendations. These results are delivered to the farmer through an intuitive interface, ensuring clarity and usability. The system can also be updated continuously with new data to improve accuracy and adaptability.

Chapter 4

REQUIREMENT SPECIFICATIONS

4.1 Hardware Requirements

- Processor: Intel i5 or higher
- RAM: Minimum 8 GB
- Storage: 256 GB
- Internet Connection: Required for dataset access, APIs, and deployment

4.2 Software Requirements

- Operating System: Windows 10/11 or Linux
- Programming Language: Python 3.8 or above
- Machine Learning Libraries: Scikit-learn, XGBoost, SHAP, LIME
- Frontend Technologies: React.js, HTML, CSS, JavaScript
- Backend Framework: Flask / Django
- Database: Firebase / MongoDB / MySQL

4.3 Functional Requirements

- Data collection from soil, weather, and agricultural datasets
- Data preprocessing and feature engineering
- Machine Learning–based crop recommendation system
- Irrigation and fertilizer recommendation module
- Explainable AI (SHAP & LIME) for prediction explanation
- Interactive web dashboard for visualization and advisory

4.4 Non-Functional Requirements

- High prediction accuracy and reliability
- User-friendly interface for farmers
- Real-time or near real-time data processing
- Secure data handling and storage
- Scalability to support multiple users and datasets
- Compatibility across desktop and mobile devices

Chapter 5

REQUIREMENT ANALYSIS

5.1 Use Case Diagram (UML)

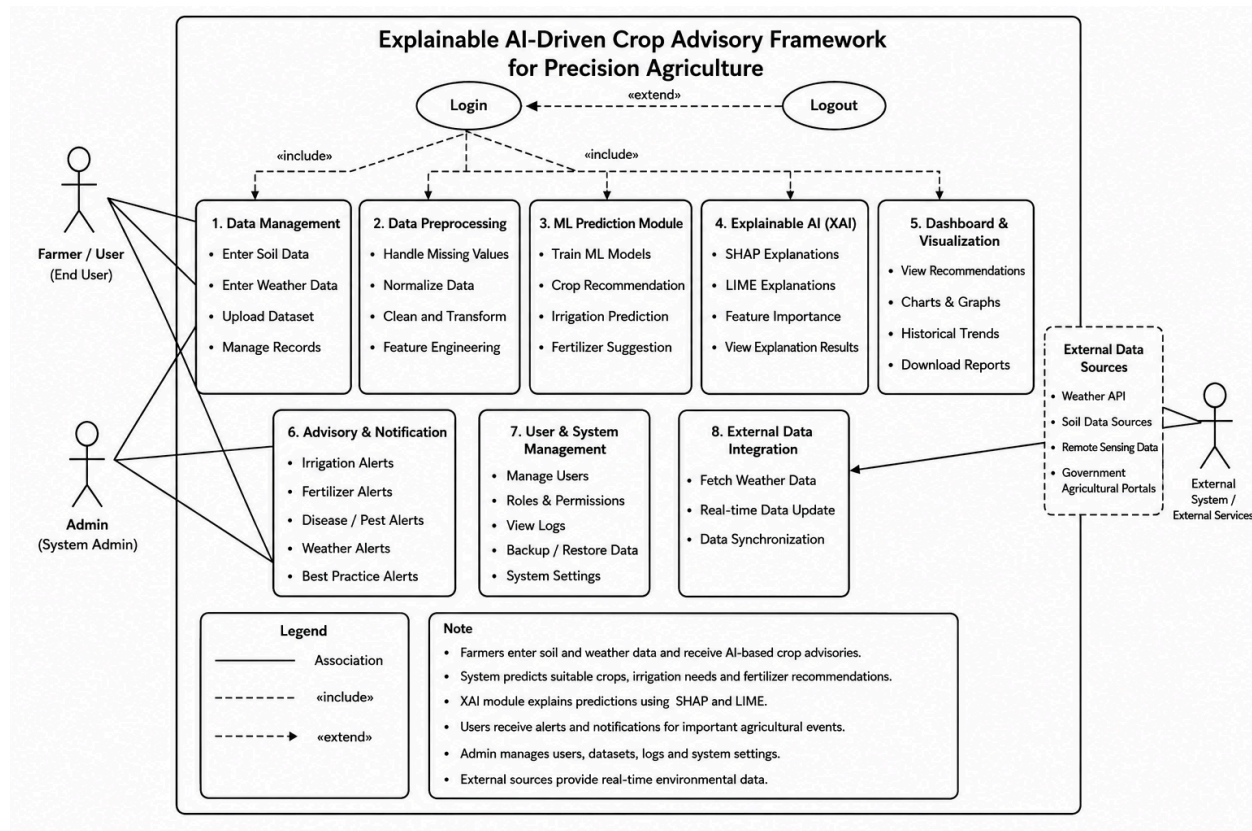


Fig 5.1 : Use Case diagram for Explainable AI-Driven Crop Advisory Framework for Precision Agriculture

The Use Case Diagram of the **Explainable AI-Driven Crop Advisory Framework for Precision Agriculture** illustrates the interaction between the Farmer/User, Admin, and External System Services with different system modules. Farmers can enter soil and weather data, upload datasets, generate crop recommendations, irrigation predictions, fertilizer suggestions, and view XAI outputs, dashboards, reports, and alerts. The Admin manages users, datasets, permissions, and system settings, while external services provide real-time environmental data for accurate crop advisory and monitoring.

5.2 Data Flow Diagram (DFD)

Data flow Diagram(Level 0):

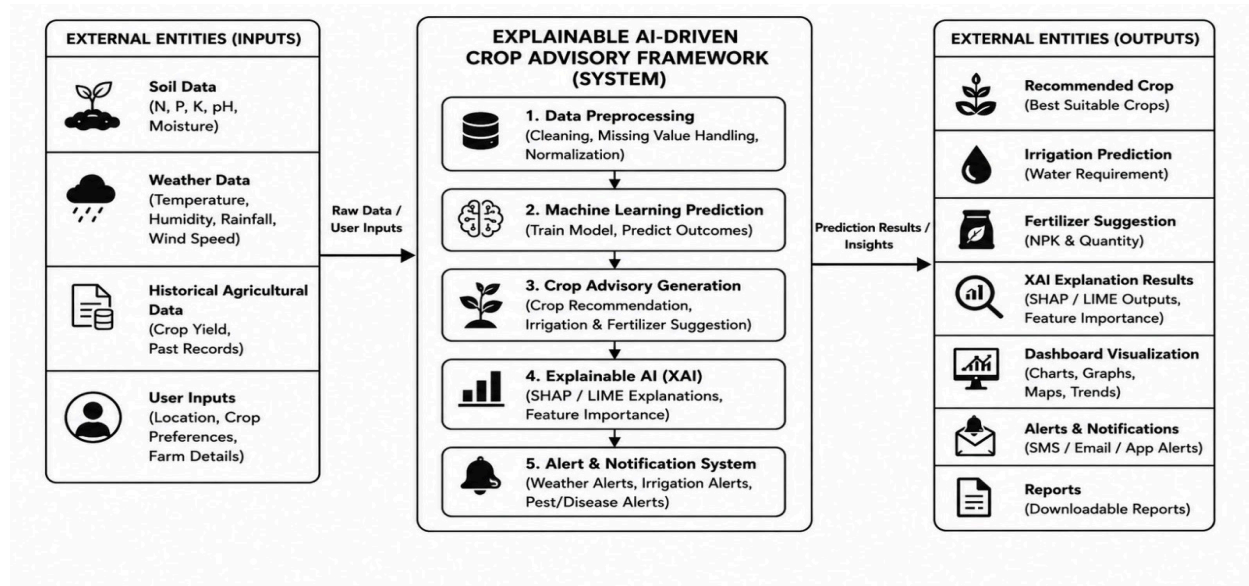


Fig 5.2.1 : Level 0 DFD-Workflow for Explainable AI-Driven Crop Advisory Framework for Precision Agriculture

The Level 0 The Data Flow Diagram (DFD) of the **Explainable AI-Driven Crop Advisory Framework for Precision Agriculture** illustrates the flow of agricultural data from input to prediction and advisory generation. The system accepts soil data, weather data, historical agricultural records, and user inputs for preprocessing and Machine Learning prediction. It generates crop recommendations, irrigation predictions, fertilizer suggestions, XAI outputs, dashboard visualizations, alerts, and reports for effective agricultural decision-making. The diagram provides a high-level overview of how data is processed and transformed into intelligent crop advisory results.

The Level 0 DFD represents the major functional components of the system and the interaction between input data, processing modules, and output results. It helps in understanding how the system collects agricultural information, processes it using AI and Machine Learning techniques, and delivers accurate recommendations and insights to farmers through a user-friendly advisory platform.

Data Flow Diagram (Level 1)

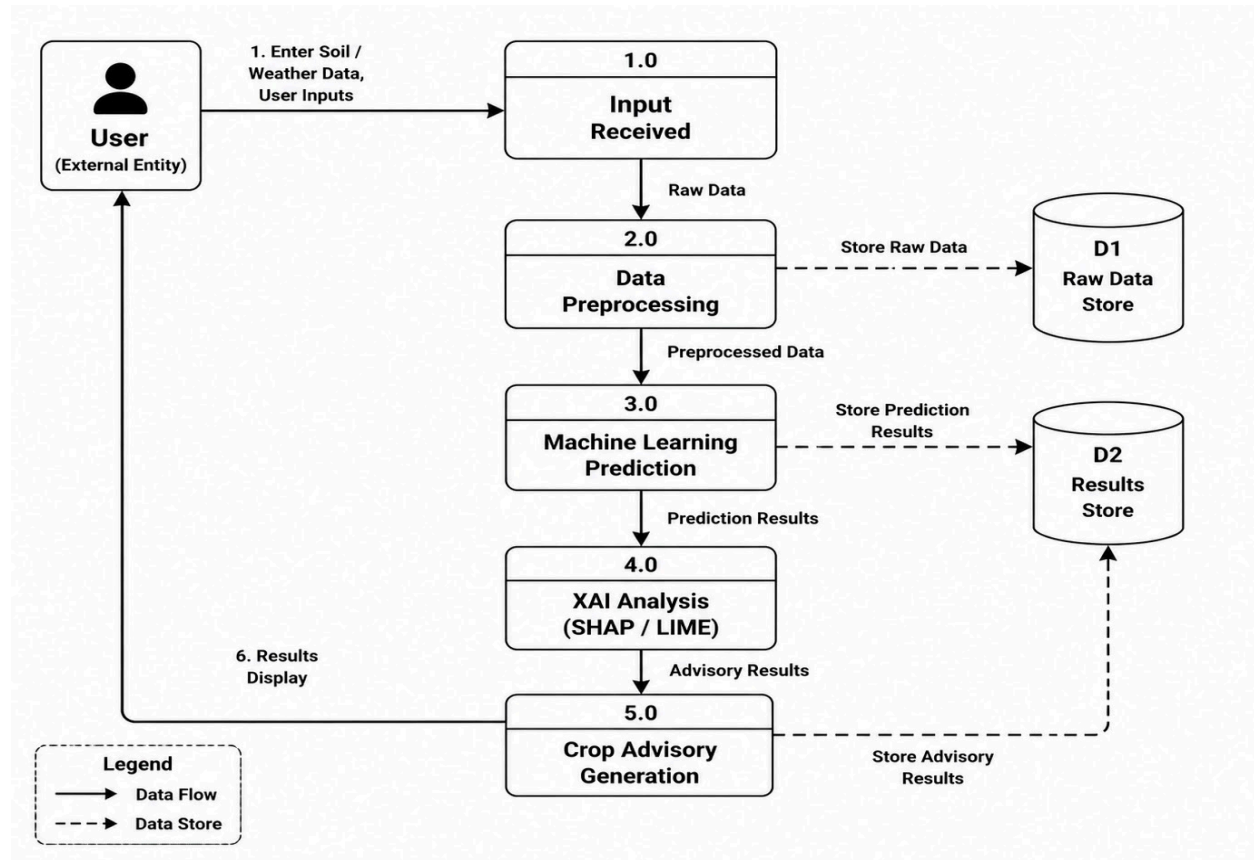


Fig 5.2.2: Level 1 DFD-Workflow for Explainable AI-Driven Crop Advisory Framework for Precision Agriculture

The Level 1 Data Flow Diagram (DFD) of the Explainable AI-Driven Crop Advisory Framework for Precision Agriculture explains the internal flow of data and processing stages within the system. The process begins with the user entering soil, weather, and agricultural data, which is received and stored in the raw data repository. The system then performs data preprocessing such as cleaning, normalization, and feature engineering before applying Machine Learning models for crop recommendation, irrigation prediction, and fertilizer suggestion. The prediction results are further analyzed using Explainable AI techniques such as SHAP and LIME to provide transparent and interpretable outputs. Finally, the system generates crop advisory results and stores prediction history and reports in the results database for future reference and visualization.

5.3 Entity-Relationship (ER) Diagram

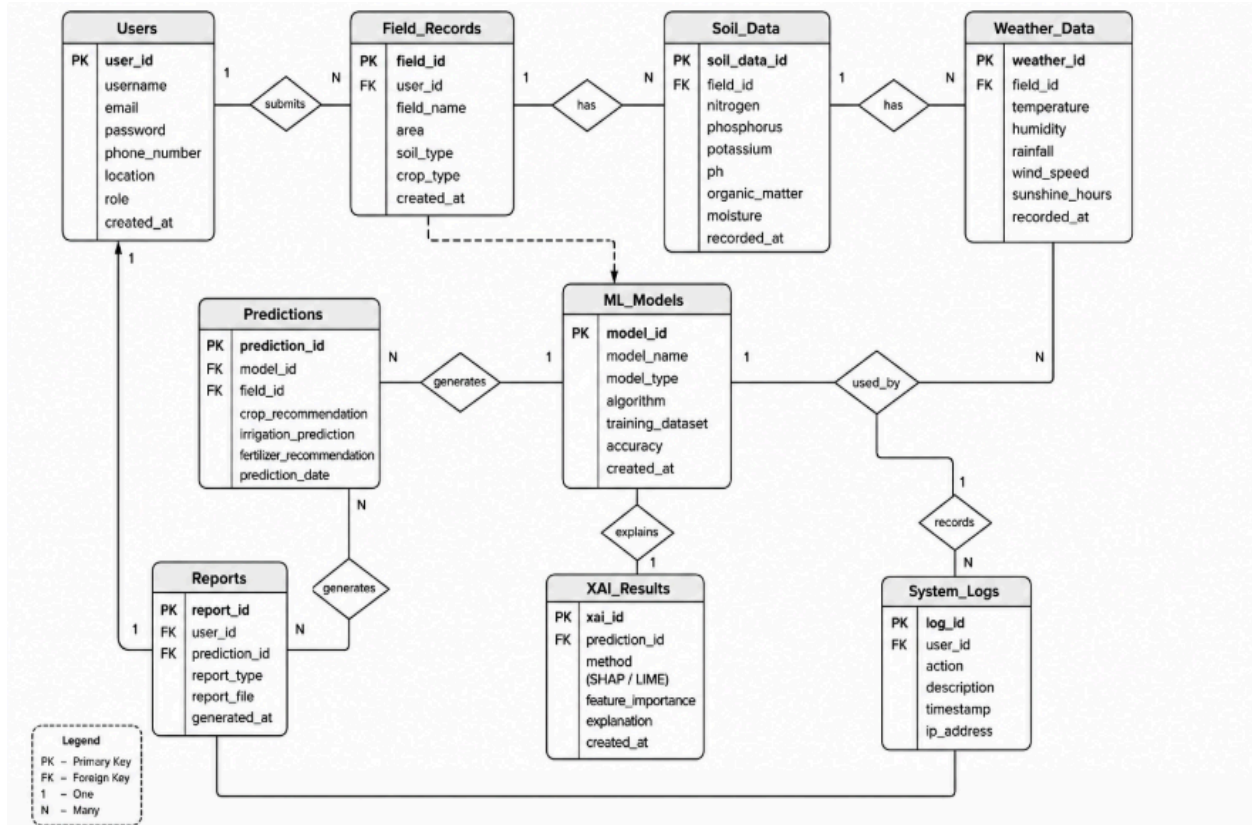


Fig 5.3 : Entity Relationship for Explainable AI-Driven Crop Advisory Framework for Precision Agriculture

The Entity-Relationship (ER) Diagram of the Explainable AI-Driven Crop Advisory Framework for Precision Agriculture defines the database structure and relationships between entities used in the system. The diagram includes entities such as Users, Field Records, Soil Data, Weather Data, ML Models, Predictions, XAI Results, Reports, and System Logs. Users submit agricultural field records containing soil and weather information, which are processed by Machine Learning models to generate crop recommendations, irrigation predictions, and fertilizer suggestions. The system also stores Explainable AI (XAI) outputs using SHAP and LIME for transparent prediction analysis. Reports, prediction history, and system logs are maintained in the database for monitoring, visualization, and future reference.

5.4 Workflow / Pipeline Diagram

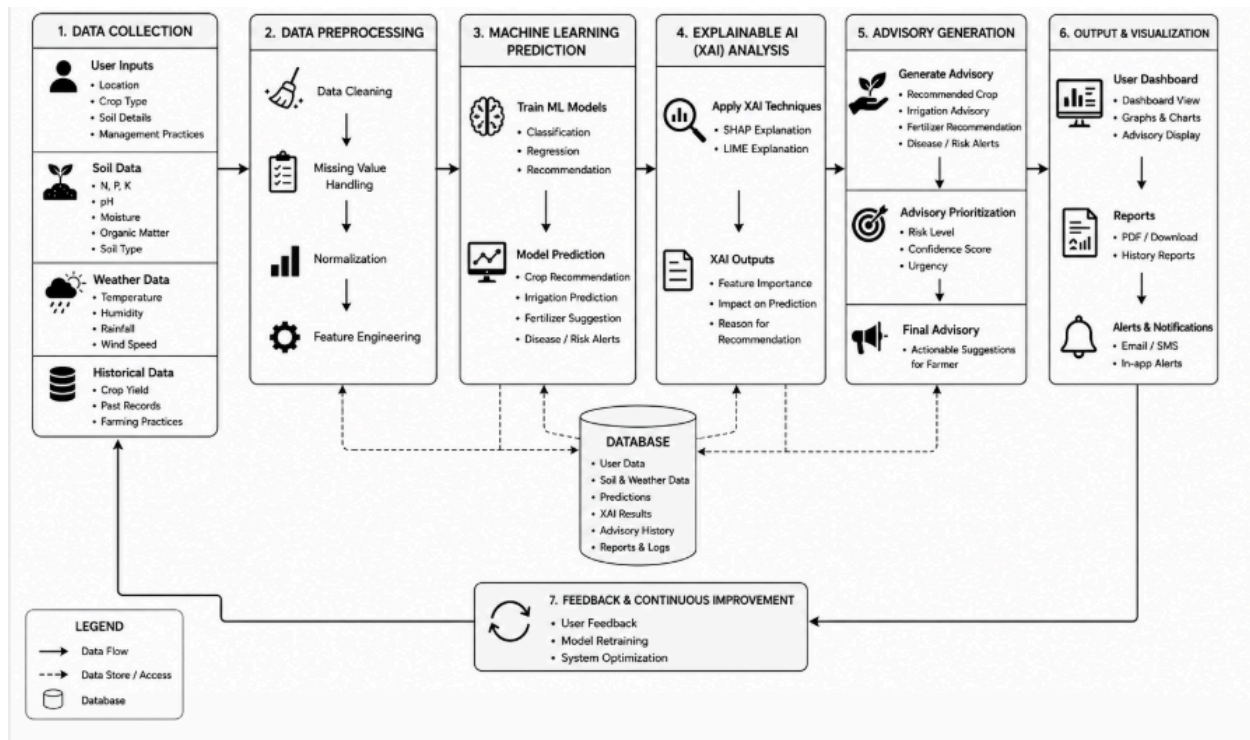


Fig 5.4 : Workflow Diagram for Explainable AI-Driven Crop Advisory Framework for Precision Agriculture

1. Data Collection

- Users enter soil, weather, and agricultural data.
- Supports manual input and dataset upload.
- Input data is validated before processing.

2. Data Preprocessing

- Cleans and normalizes agricultural data.
- Handles missing values and noisy data.
- Performs feature engineering and transformation.

3. Machine Learning Prediction

- Trains Machine Learning models using datasets.
- Generates crop recommendation predictions.
- Predicts irrigation and fertilizer requirements.
- Computes prediction accuracy and confidence scores.

4. Explainable AI (XAI) Analysis

- Applies SHAP and LIME explanation techniques.
- Identifies important prediction features.
- Provides transparent and interpretable outputs.

5. Advisory Generation

- Generates crop advisory recommendations.
- Produces irrigation and fertilizer suggestions.
- Generates disease and risk alerts.

6. Output & Visualization

- Displays prediction results to users.
- Shows charts, graphs, and dashboard analytics.
- Allows report generation and result export.

Chapter 6

SYSTEM ARCHITECTURE

6.1 Overall System Architecture

System architecture defines the structural design of the Explainable AI-Driven Crop Advisory Framework for Precision Agriculture. The architecture describes how different modules interact with each other to collect agricultural data, preprocess datasets, train Machine Learning models, generate crop advisory predictions, and provide Explainable AI (XAI) outputs.

The proposed architecture follows a modular design approach consisting of data collection, data preprocessing, machine learning prediction, explainable AI analysis, advisory generation, dashboard visualization, and database management components.

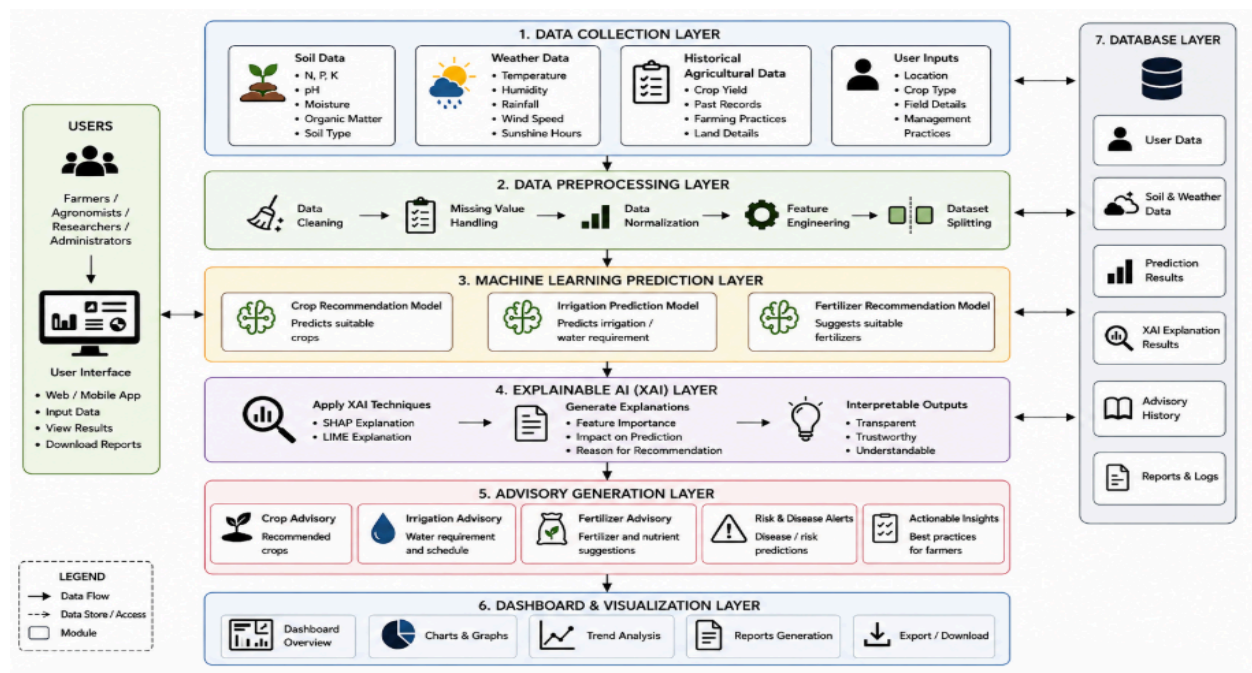


Fig 6.1 : Overall System Architecture for Explainable AI-Driven Crop Advisory Framework for Precision Agriculture

1. Data Collection Layer

This module is responsible for collecting datasets related to:

- Soil Data
- Weather Data
- Historical Agricultural Data
- Crop Yield Records
- User Inputs
- Farming Practices

2. Data Preprocessing Layer

- Data Cleaning – Removes noisy and inconsistent agricultural records.
- Missing Value Handling – Fills or removes missing values from datasets.
- Data Normalization – Converts values into a common scale for processing.
- Feature Engineering – Extracts important features for prediction models.
- Dataset Splitting – Divides data into training and testing datasets.
- Data Transformation – Converts raw agricultural data into structured format.

3. Machine Learning Prediction Layer

1. Crop Recommendation Model

- Predicts suitable crops based on soil and weather conditions.
- Uses Machine Learning algorithms for accurate prediction.
- Improves crop productivity and yield estimation.

2.Irrigation Prediction Model

- Predicts crop water requirements.
- Helps optimize irrigation scheduling.
- Reduces water wastage in farming practices.

3.Fertilizer Recommendation Model

- Suggests suitable fertilizers and nutrient quantities.
- Improves soil fertility and crop health.
- Generates fertilizer advisory recommendations.

4. Explainable AI (XAI) Layer

The Explainable AI module performs the following tasks:

- Applies SHAP and LIME explanation techniques
- Identifies important prediction features
- Generates transparent prediction explanations
- Improves trust and interpretability of AI outputs
- Provides reasoning behind crop recommendations

5. Advisory Generation Layer

- Crop recommendation advisory
- Irrigation scheduling recommendations
- Fertilizer suggestion generation
- Disease and risk alert generation
- Actionable farming insights for users

6.Dashboard & Visualization Layer

- Responsive dashboard interface

- Interactive charts and graphs
- Prediction result visualization
- Trend analysis reports
- Downloadable reports and analytics
- Alert and notification display

7. Database Layer

- User details
- Soil and weather datasets
- Historical agricultural records
- Prediction results
- XAI explanation outputs
- Reports and system logs

Chapter 7

DESIGN DETAILS

Design details describe the internal implementation structure of the proposed Explainable AI-Driven Crop Advisory Framework for Precision Agriculture. The design phase focuses on data collection, preprocessing techniques, Machine Learning model implementation, Explainable AI analysis, advisory generation, dashboard visualization, and database management.

7.1 System Design

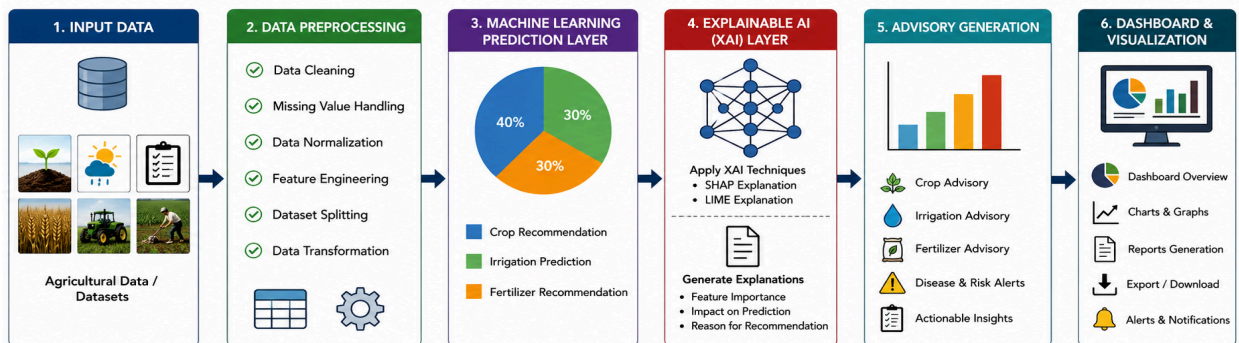


Fig 7.1 : Architecture Flow for Explainable AI-Driven Crop Advisory Framework for Precision Agriculture

The architecture flow diagram illustrates the working process of the proposed AI-based crop advisory system. Initially, the system collects agricultural input datasets such as soil data, weather data, historical agricultural records, and user inputs. The collected data then undergoes preprocessing operations including data cleaning, missing value handling, normalization, feature engineering, and dataset transformation to improve data quality and model performance.

After preprocessing, the processed datasets are used by Machine Learning models to generate crop recommendations, irrigation predictions, and fertilizer suggestions. Explainable AI techniques such as SHAP and LIME are applied to provide transparent prediction explanations and feature importance analysis. Finally, the system generates crop advisories, risk alerts, dashboard visualizations, reports, and downloadable analytics for farmers and agricultural users. This workflow improves prediction accuracy, interpretability, and efficient agricultural decision-making.

7.2 Sequence Diagram

The sequence diagram describes the interaction between the farmer/user and the different modules of the **Explainable AI-Driven Crop Advisory Framework for Precision Agriculture**. It explains how the system processes agricultural data, generates Machine Learning predictions, applies Explainable AI techniques, and provides advisory recommendations through the dashboard interface.

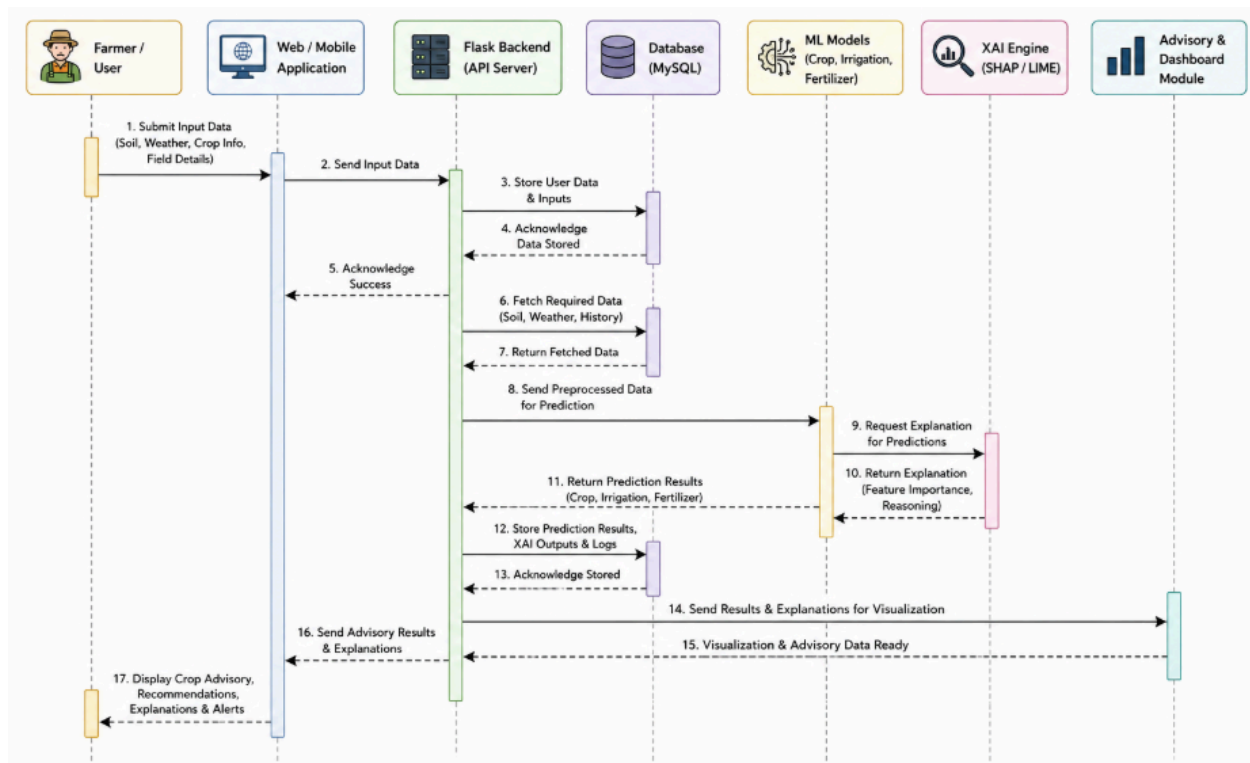


Fig 7.2 : Sequence Diagram for Explainable AI-Driven Crop Advisory Framework for Precision Agriculture

The working process is as follows:

- **User Interaction:** The farmer/user submits soil, weather, crop, and field details through the web/mobile application.
- **Data Handling:** The Flask backend stores the user data in the database and fetches required historical agricultural datasets.

- Data Preprocessing: The system cleans, normalizes, and transforms the collected agricultural data before prediction.
- Prediction Process: The Machine Learning models generate crop recommendations, irrigation predictions, and fertilizer suggestions.
- Explainable AI: The XAI engine generates explanations using SHAP and LIME techniques to improve interpretability and transparency.
- Result Generation: Prediction outputs and explanation results are stored in the database for future analysis and reporting.
- Advisory & Visualization: The system sends advisory results, charts, reports, and alerts to the dashboard module for visualization.
- Final Output: The dashboard displays crop advisory recommendations, irrigation schedules, fertilizer suggestions, explanations, and alerts to the user.

7.3 CNN Architecture

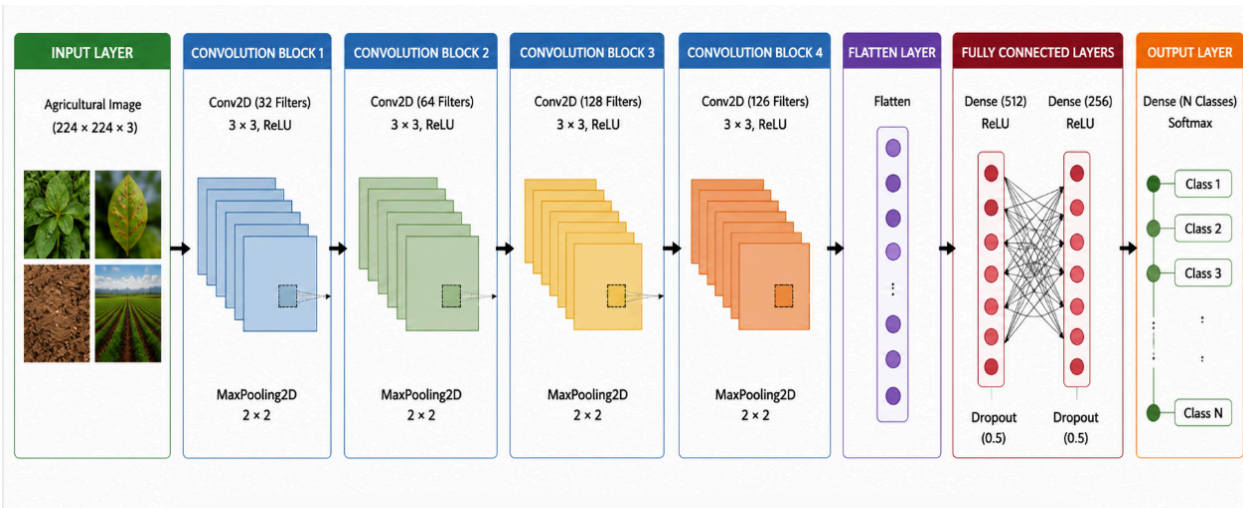


Fig 7.3 : CNN Architecture for Explainable AI-Driven Crop Advisory Framework for Precision Agriculture

1. Input Layer

- Agricultural images such as leaf images, soil images, and crop field images are given as input to the CNN model.
- Input image size is typically $224 \times 224 \times 3$.
- The input contains RGB color channels for feature extraction.

2. Convolution Block 1

- Uses 32 filters with kernel size 3×3 .
- Applies ReLU activation function.
- Extracts basic features such as edges, textures, and patterns.
- MaxPooling layer with 2×2 pooling reduces image dimensions.

3. Convolution Block 2

- Uses 64 filters with kernel size 3×3 .
- Extracts more detailed agricultural image features.
- Improves learning of crop and soil patterns.
- MaxPooling layer further reduces feature map size.

4. Convolution Block 3

- Uses 128 filters with kernel size 3×3 .
- Extracts high-level features from agricultural images.
- Improves feature learning for crop disease and soil analysis.
- MaxPooling layer improves computational efficiency.

5. Convolution Block 4

- Uses 256 filters with kernel size 3×3 .
- Learn complex image features and hidden patterns.
- Enhances prediction accuracy for agricultural analysis.
- MaxPooling reduces unnecessary feature dimensions.

6. Flatten Layer

- Converts 2D feature maps into a 1D feature vector.
- Prepares extracted features for dense layer processing.
- Connects convolution layers to fully connected layers.

7. Fully Connected Layers

- Dense layers contain 512 and 256 neurons.
- Uses ReLU activation function.
- Performs classification and feature learning.
- Dropout layers reduce overfitting and improve model performance.

8. Output Layer

- Uses Softmax activation function for multi-class classification.
- Generates final prediction outputs.
- Predicts crop condition, disease category, or agricultural class labels.
- Produces probability scores for each output class.

Chapter 8

CONCLUSION

8.1 Conclusion

Explainable AI-Driven Crop Advisory Framework for Precision Agriculture mainly focused on the analysis, design, and planning of an intelligent crop advisory system using Machine Learning and Explainable AI techniques. During this phase, the problem statement, objectives, literature survey, methodology, requirement analysis, system architecture, UML diagrams, DFDs, ER diagrams, and workflow design were successfully completed. The proposed framework was designed to provide crop recommendations, irrigation suggestions, fertilizer advisory, and transparent prediction explanations using SHAP and LIME techniques. The study of existing systems helped in identifying the limitations of traditional black-box agricultural models and the need for interpretable AI solutions. Overall, Phase 1 established a strong foundation for developing an efficient, user-friendly, and intelligent precision agriculture system in the next phase of the project.

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